Arora V Analog to Digital Converter (ADC)

User Guide

UG299-1.0.6E, 2025-12-18

Revision History

Date	Version	Description
5/08/2023	1.0E	Initial version published.
7/18/2023	1.0.1	ADC conversion timing description optimized.
12/08/2023	1.0.2	ADC Parameter Configuration description optimized.
2/02/2024	1.0.3	• ADC input description optimized. • Table 3-3 ADC Configuration Interface Parameters updated.
12/30/2024	1.0.4	ADC timing Parameters and ADC electrical parameters added.Table 1Table 1
4/29/2025	1.0.5	• *5 ADC (60K)* added. • *4 ADC (75K/138K)* And *3 ADC(25K)* refined.
12/18/2025	1.0.6	*mDRP Control Registers* updated.

Contents

List of Figures

List of Tables

1 About This Guide

1.1 Purpose

Arora V Analog to Digital Converter (ADC) User Guide is to help you quickly learn the features and usage of Arora V ADC by introducing to the functions, ports, configuration, etc.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI website. You can find the related documents at *www.gowinsemi.com*:

- *DS981, GW5AT series of FPGA Products Data Sheet*
- *DS1103, GW5A series of FPGA Products Data Sheet*
- *DS1239, GW5AST series of FPGA Products Data Sheet*
- *DS1105, GW5AS series of FPGA Products Data Sheet*
- *DS1108, GW5AR series of FPGA Products Data Sheet*
- *DS1118, GW5ART series of FPGA Products Data Sheet*
- *SUG100, Gowin Software User Guide*

1.3 Terminology and Abbreviations

The terminology and abbreviations used in this manual are as shown in Table 1-1.

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Full Name	Meaning
ADC	Analog to Digital Converter	-
CIC Filter	Cascaded Integrator-comb Filter	-
FPGA	Field Programmable Gate Array	-
IP	Intellectual Property	-
OSC	Oscillator	-

Terminology and Abbreviations	Full Name	Meaning
SRAM	Static Random Access Memory	-
ADCULC	-	ADC Location: Top Left
ADCLRC	-	ADC Location: Bottom Right

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: *www.gowinsemi.com*

E-mail：*support@gowinsemi.com*

2 Overview

Arora V FPGA products integrate eight-channel 10 bits Delta-sigma ADC. It is an ADC with low power, low-leakage current, and high dynamic performance. Combined with the programmable logic capability of the FPGA, and the integrated voltage and temperature sensing unit, the ADC can meet the internal temperature and power monitoring and data collection requirements. At the same time, the FPGA provides rich and free configurable GPIO differential interfaces and ADC analog differential signal interfaces to connect to the voltage channel of the ADC, which can meet the requirements for external voltage data collection and monitoring.

2.1 Features

Arora V ADC main features are as follows:

- Number of ADC cores:
  - GW5A-25/ GW5AR-25/ GW5AS-25: 1
  - GW5A-138/ GW5AT-138/ GW5AT-75/ GW5AST-138: 2
  - GW5A-60/ GW5AT-60: 1 delta-sigma ADC, 1 SARADC[1]

    Note!

    [1] For more information about the SARADC, please refer to *UG298E, Arora V SAR ADC User Guide*.
  - GW5AT-15/ GW5ART-15: 1
- Reference voltage source: Built-in
- Number of channels per ADC: 8
- Bit width accuracy: 10 bits
- Sampling Clock: < 10MHz
- Differential input voltage range: 0 V ~ 1 V (Positive input voltage > Negative input voltage).
- Temperature sensor accuracy: +/-2℃

- Voltage sensor accuracy: +/-5mV

2.2 Functional Description

2.2.1 Overview

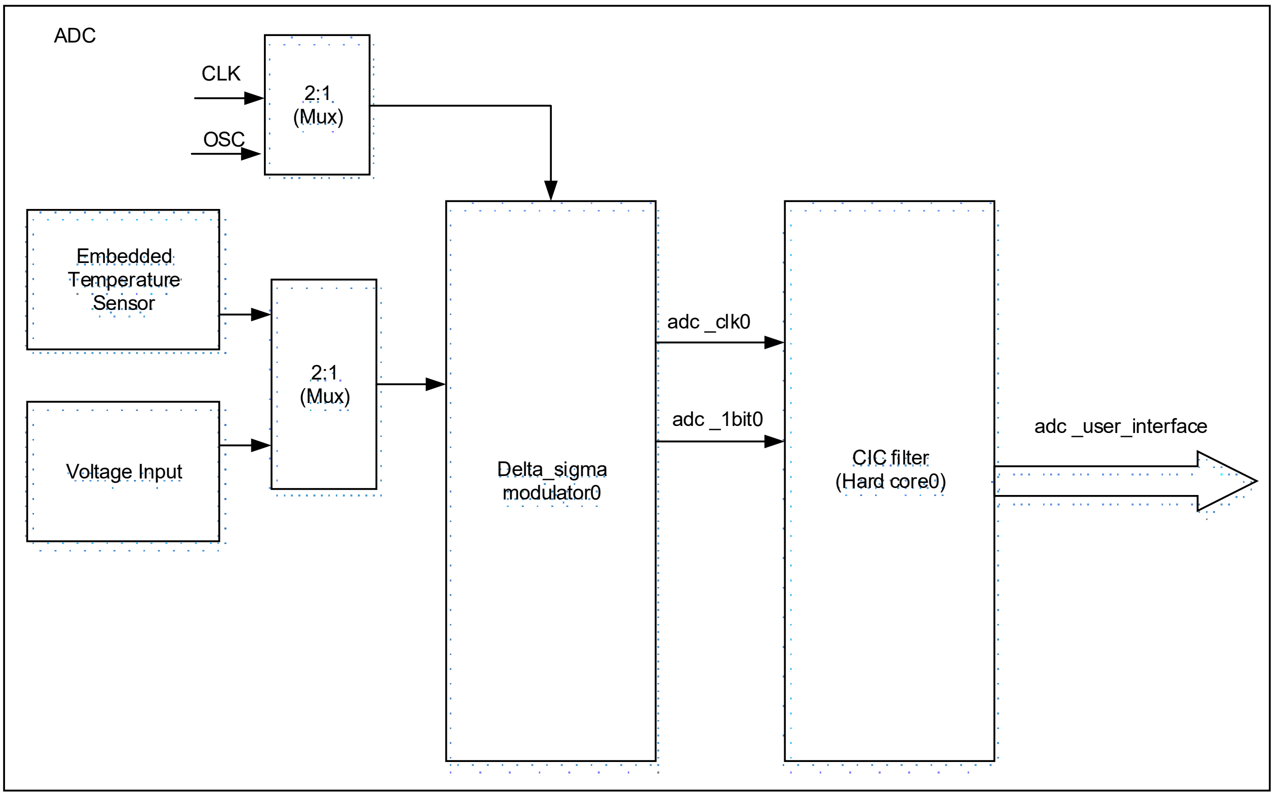
Arora V ADC provides analog Delta-sigma modulators to meet the requirements of on-chip temperature and voltage detection in multiple regions, and also provides abundant input interfaces to meet off-chip voltage and temperature input, supporting single-ended and differential signal input. (Positive input voltage > Negative input voltage)

Arora V ADC has embedded reference voltage source with high accuracy, and it does not require off-chip voltage reference source; Arora V ADC features low power and high accuracy for temperature and supply voltage detection. Arora V ADC has an internally integrated voltage signal processing module, so no external voltage reference source is required. It meets the accuracy of voltage signal measurements and helps to reduce user cost.

2.2.2 Architecture Diagram

Figure 2-1 shows the architecture diagram of Arora V ADC.

Figure 2-1 ADC Diagram



Arora V ADC supports both on-chip temperature sensing and voltage detection modes. By configuring the control signals, users can select the voltage output from the internal temperature sensor to enter the temperature detection mode. Alternatively, users can select another measurement path to monitor the voltages of FPGA internal IP modules, Bank voltages, core voltage, SRAM voltage, etc. External voltage signals can be routed to the ADC through differential GPIO pads or dedicated ADC input pads within the IO Banks for quantization. Avoid using digital IOs in the same Bank as the ADC inputs to minimize noise.

The Arora V ADC allows selection between the CLK input (provided by User Logic) and an internal OSC clock source. By choosing the appropriate clock source, users can achieve an optimal balance between power consumption and performance.

The voltage signal into the Delta_sigma modulator0 is quantized and noise-shaped to output adc_1bit0 and adc_clk0, which can be sent to the embedded CIC hard core or CIC soft core for further processing to obtain the digital characterization of temperature and voltage.

Additionally, certain packages provide two pairs of dedicated differential ADC input interfaces: adcvp/adcvn and adctp/adctn, offering users low-latency and low-noise differential voltage input channels.

2.3 ADC Characteristics

2.3.1 ADC Conversion Timing

There are N clock cycles needed for ADC to sample analog input signals and convert them to output digital signals; then output signals are generated. When the rising edge of the ADC sampling request signal sensor_req occurs while the ADC enable signal sensor_en is active (high level effective), it triggers the ADC to perform a sampling process. Once the number of sampled points reaches the set value, the sensor_rdy signal will be pulled high to indicate the completion of sampling, and the sampled value sensor_value [13:0] will be output.

In voltage measurement mode: The sensor_value [13:0] is a 14-bit unsigned value, which should be divided by 2048 and then multiplied by the voltage divider ratio to obtain the actual measured value in volts (V).

In temperature mode, sensor_value is a signed number (sensor_value[13] represents the sign bit, and sensor_value[12:0] represent the data bits). The data bits must be divided by 4 to obtain the actual measured value, with the unit in °C.

Figure 2-2 ADC Conversion Timing

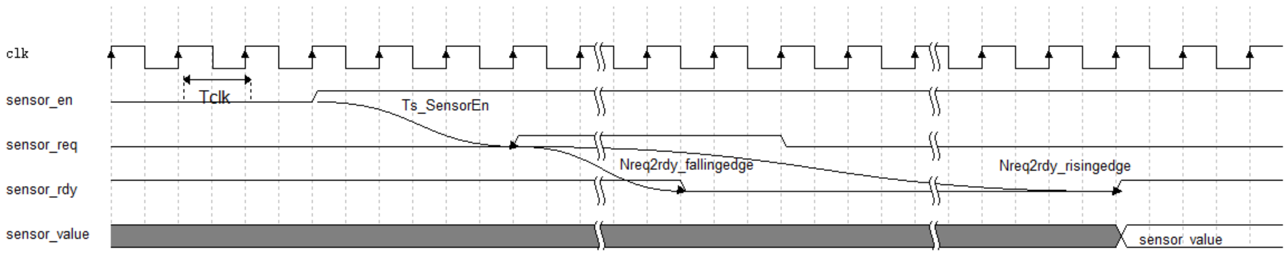


Table 2-1 ADC Timing Parameters

Symbol	Description	Spec.		Unit
		Min.	Max.	
Tclk	Clock cycle	100	-	ns
$T_{S\_SensorEn}$	Setup Time from SensorEn rising edge to Sensor Req	10	-	ns
$N_{req2rdy\_fallingedge}$	Clock cycles (SensorReq rising edge to Sensor_ready falling edge)	-	10	Cycle
$N_{req2rdy\_risingedge}$	Clock cycles (SensorReq rising edge to Sensor_ready falling edge)	128	2048	Cycle

2.3.2 Electrical Characteristic Parameters

Table 2-2 ADC Electrical Parameters

Parameter	Description	Spec.			Unit
		Min.	Typ.	Max.	
DC precision					
Output	Digital output bits	-	14	-	Bit
INL	Integral nonlinearity	-2	-	2	LSB
DNL	Differential nonlinearity	-1	-	1	LSB
Analog Input					
Vref	Internal voltage reference		1.0	-	V
CH[7:0]	Single-ended input range	0	-	1	V
CIN	Input capacitance	-	1.3	-	pF
Slew Rate					
Fs	Ratio	-	-	10	MHz
$N_{req2rdy\_risingedge}$	Conversion cycles	128	-	2048	Clock cycle
Dynamic Characteristic Parameters					
SNRFS[2]	Signal Noise Ratio	-	60	-	dB
ENOB	Valid output data bits	-	9.6	-	Bit
V_{INBW}	Dedicated input pin -3dB bandwidth	-	5	-	MHz
Supply voltage[3]					
V_{dd_a}	Analog core voltage	-	0.9	-	V

Note!

- [1] Dedicated-pin capacitor
- [2] Test Condition Vin= -0.5dBFs, f_{IN}=1KHz。
- [3] For the power supply voltage range, please refer to the relevant device datasheet.

3 ADC(25K)

3.1 Devices Supported

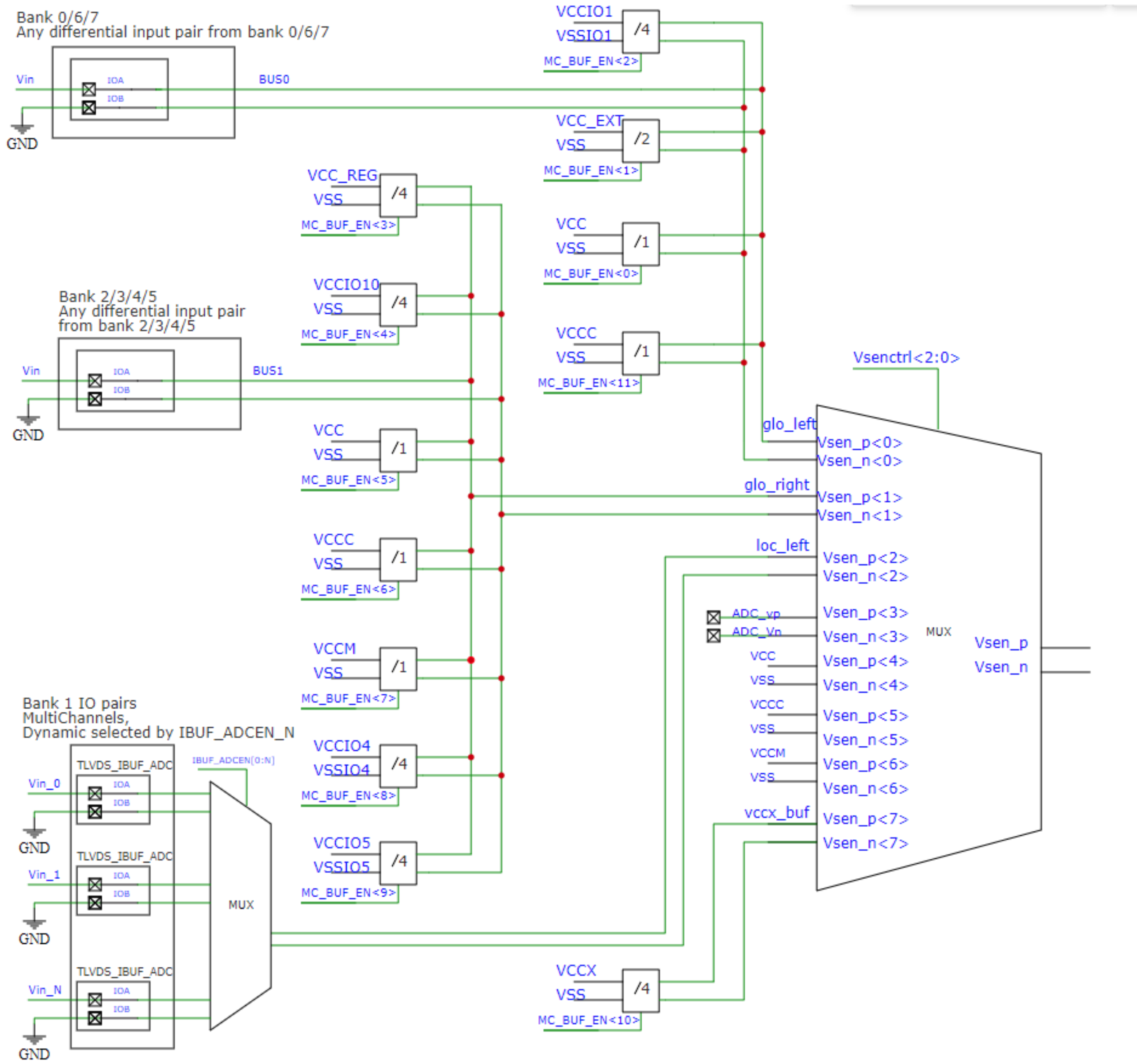
Table 3-1 Devices Supported

Product Family	Series	Device
Arora	GW5A	GW5A-25A
	GW5AR	GW5AR-25A
	GW5AS	GW5AS-25A

3.2 Input Channel Selection (Voltage Divider Ratio)

The input channel selection diagram for voltage mode ADC in GW5A-25 / GW5AR-25 / GW5AS-25 devices is shown below.

Figure 3-1 ADC Input Channel Selection Diagram in Voltage Mode



The voltage mode ADC input channel selection and voltage divider ratio table for GW5A-25 / GW5AR-25 / GW5AS-25 is shown below.

Table 3-2 ADC Input Channel Selection and Divider Ratio in Voltage Mode

Vsenctl[2:0]	ADC Bus	Input Channel	Divider Ratio	BUF_EN[11:0]	Note
000	glo_left	VCC	1	BUF_EN[0]	
		VDD12_MIPI	2	BUF_EN[1]	VCC_EXT

Vsenctl[2:0]	ADC Bus	Input Channel	Divider Ratio	BUF_EN[11:0]	Note
000	glo_left	VCCIO1	4	BUF_EN[2]	
		PAD	1		BANK0/6/7 IO
001	glo_right	VCC_LDO	4	BUF_EN[3]	VCC_REG
		VCCIO10	4	BUF_EN[4]	
		VCC	1	BUF_EN[5]	
		VCCM	1	BUF_EN[7]	
		VCCIO4	4	BUF_EN[8]	
		VCCIO5	4	BUF_EN[9]	
		PAD	1		BANK2/3/4/5 IO
010	loc_left		1		BANK1 IO
011	adcv		1		
100	vcc		1		
101	vccc		1		
110	vccm		1		
111	vccx_buf	VCCX	4	BUF_EN[10]	

3.3 ADC Instantiation

3.3.1 Port Diagram

Figure 3-2 ADC Port Diagram (Static Input Channel)

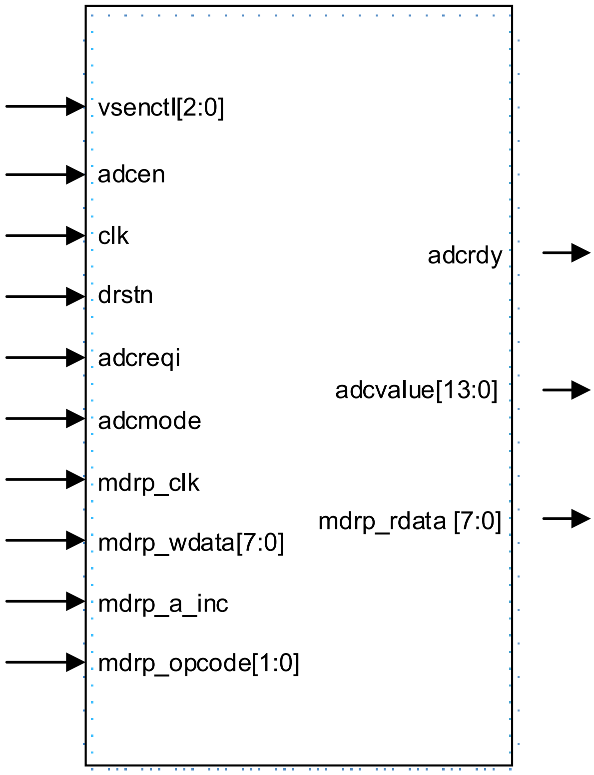
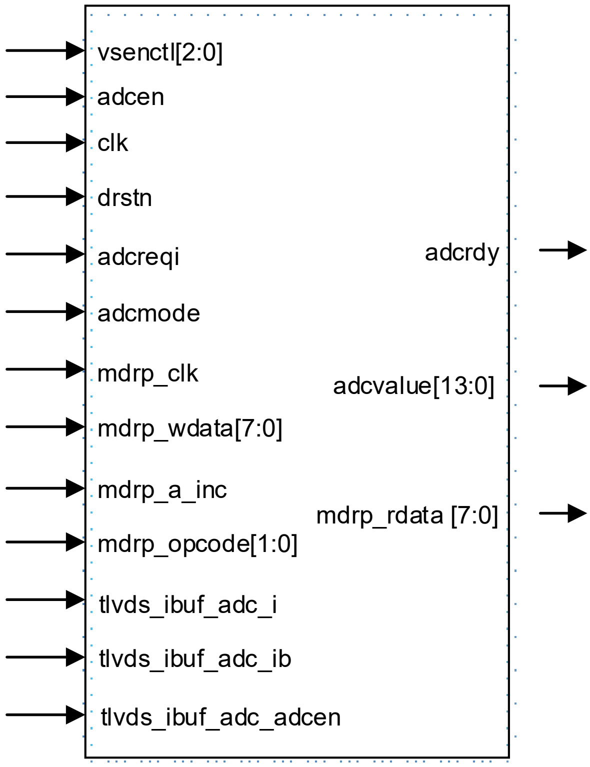


Figure 3-3 ADC Port Diagram (Dynamic Input Channel)



3.3.2 Port Description

ADC Port

Table 3-3 ADC Port Description

Port	I/O	Description
vsenctl[2:0]	input	Input source selection bit [2:0] ● 3'b000: glo_left ● 3'b001: glo_right ● 3'b010: loc_left (Corresponding to Bank1 GPIO) ● 3'b011: adcv ● 3'b100: vcc ● 3'b101: vccc ● 3'b110: vccm ● 3'b111: vccx_buf
adcen	input	Enable signal, active high
clk	input	Clk input
drstn	input	Digital part reset signal, active low
adcreqi	input	Measurement request signal, valid rising edge
adcmode	input	Mode selection

Port	I/O	Description
adcmode	input	● 1’b0: temperature mode ● 1’b1: voltage mode
adcrdy	output	Measurement completion signal, active high
adcvalue	output	bit[13:0] the measurement result output

ADC_mDRP Port

Table 3-4 ADC_mDRP Port Description

Port	I/O	Description
mdrp_clk	input	mdrp clock
mdrp_wdata[7:0]	input	bit[7:0] mdrp_wdata
mdrp_a_inc	input	mdrp_a_inc
mdrp_opcode[1:0]	input	bit[1:0] mdrp_opcode
mdrp_rdata	output	bit[7:0] mdrp_rdata

mDRP Control Registers

Table 3-5 ADC mDRP Control Registers

Address	Register name	R/W	Description
0x00	sensor_adc_ctrl1	RW	Bit[7:6] reserved Bit[5:3] cfg_sample_cnt_sel ● 001:128 ● 010:256 ● 011:512 ● 100:1024 Bit[2:0] gain_ctrl: Keep at 0.
0x01	sensor_adc_ctrl2	RW	Bit[7:4] reserved Bit[3] cfg_sensor_mode ● 0: Temperature ● 1:Voltage Bit[2:0] cfg_rate_change_ctrl ● 010:16 ● 011:32 ● 100:64 ● 101:128
0x02	sensor_adc_ctrl3	RW	Bit[7:0] cfg_fscal_value[7:0]
0x03	sensor_adc_ctrl4	RW	● Bit[7:2] reserved ● Bit[1:0] cfg_fscal_value[9:8]

Address	Register name	R/W	Description
0x04	sensor_adc_ctrl5	RW	Bit[7:0] cfg_offset_value[7:0]
0x05	sensor_adc_ctrl6	RW	● Bit[7:4] reserved ● Bit[3:0] cfg_offset_value[11:8]

ADC_IO_BUF Port

Table 3-6 ADC_IO_BUF Port Description

Port	I/O	Description
tlvds_ibuf_adc_i	input	The adcvp signal is input to the differential GPIO Pin A on Bank1, with Pin A connected to the voltage to be tested.
tlvds_ibuf_adc_ib	input	The adcvn signal is input to the differential GPIO Pin B on Bank1, with Pin B tied to GND.
tlvds_ibuf_adc_adcen	input	ADC enable signal for differential GPIOs; only one adcen can be enabled at a time.

Note!

Bank1 supports multiple ADC input channels.

3.3.3 Parameter Configuration

Select “Tools > IP Core Generator” in Gowin Software GUI to start the ADC configuration interface. Figure 3-4 shows the example.

Figure 3-4 ADC Configuration Interface

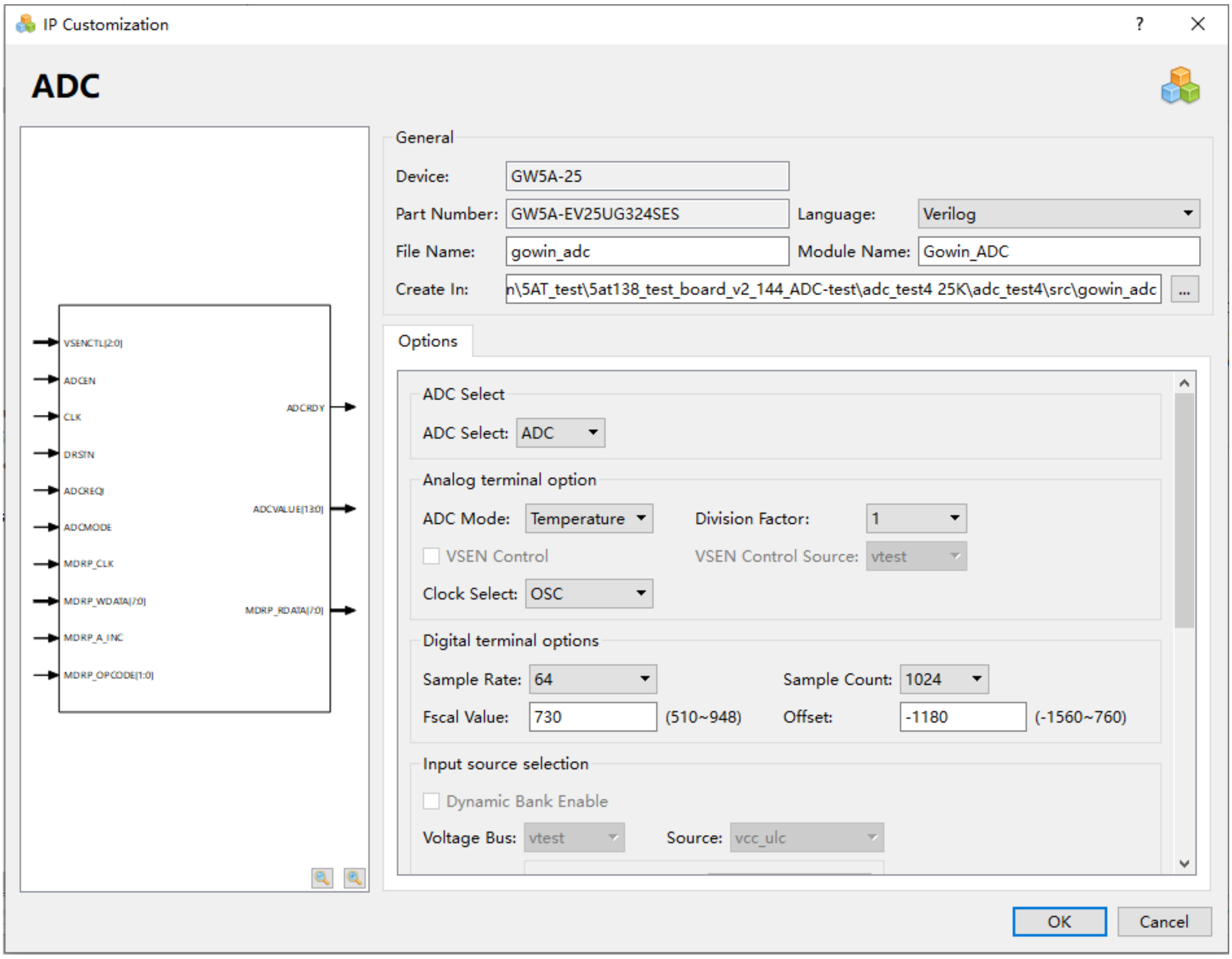


Table 3-7 ADC Configuration Parameter Description

Parameter	Default Value	Description
ADC Select	ADC	ADC
ADC Mode	Temperature	Temperature/Voltage
Division Factor	1	clock division 0: /1, 1: /2, 2: /4, 3: /8 Clock after frequency division, 500kHz~8MHz
Clock Select	OSC	clk source OSC (2.5MHz) or CLK
Sample Rate	128	sample rate configuration 16/32/64/128

Parameter	Default Value	Description
Sample Count	1024	sample count configuration 128/256/512/1024
Fscal Value	730(Temperature) 653(Voltage)	temperature mode: 510~948 voltage mode: 452~840
Offset	-1180(Temperature) 0(Voltage)	temperature mode: -1560~-760 voltage mode: -410~410
Dynamic Input Source Enable (Voltage mode)	Unchecked	When checked, the tlvds_ibuf_adc ADC input interface is enabled. (Bank1 input, supports multiple IO pairs)
glo_left (Voltage mode)	VCC	VCC/VDD12_MIPI/VCCIO1/PAD PAD corresponds to IOs in Bank0/6/7, referred to as bus0. When using IOs from these Banks as ADC inputs without selecting Dynamic Input Source Enable, the following physical constraint must be added: USE_ADC_SRC bus0 loc; loc: Specifies the physical location of the ADC input pin, e.g., IOR26. When Dynamic Input Source Enable is selected, the tlvds_ibuf_adc ADC input interface is enabled. Only one IO pair is supported. Connect terminal A to the voltage under test, and terminal B to ground (GND)..
glo_right (Voltage mode)	VCC_LDO	VCC_LDO/VCCIO10/VCC/VCCM/VCCIO4_5/PAD PAD corresponds to IOs in Bank2/3/4/5, referred to as bus1. When using IOs from these Banks as ADC inputs without selecting Dynamic Input Source Enable, the following physical constraint must be added: USE_ADC_SRC bus1 loc; loc: Specifies the physical location of the ADC input pin, e.g., IOR26. When Dynamic Input Source Enable is selected, the tlvds_ibuf_adc ADC input interface is enabled. Only one IO pair is supported. Connect terminal A to the voltage under test, and terminal B to ground (GND).
vccx_buf (Voltage mode)	VCCX	VCCX

After ADC configuration, it will generate three files that are named after the "File Name". Take the default configuration as an example:

- “gowin_adc.v” file is a complete Verilog module to generate instance Gowin_ADC;
- “gowin_adc_tmp.v” is a template file for IP designs;
- “gowin_adc.ipc” file is an IP configuration file for users to load and configure the IP.

Note!

If VHDL is selected as the hardware description language, the first two files will be named with .vhd suffix.

3.3.4 Instantiation

Verilog Instantiation:

```
Gowin_ADC Gowin_ADC_inst (
.adcrdy(adcrdy_o),
.adcvalue(adcvalue_o),
.mdrp_rdata(mdrp_rdata_o),
.vsenctl(vsenctl_i),
.adcen(adcen_i),
.clk(clk_i),
.drstn(drstn_i),
.adcreqi(adcreqi_i),
.adcmode(adcmode_i),
.mdrp_clk(mdrp_clk_i),
.mdrp_wdata(mdrp_wdata_i),
.mdrp_a_inc(mdrp_a_inc_i),
.mdrp_opcode(mdrp_opcode_i)
);
```

Vhdl Instantiation:

```
component Gowin_ADC
port (
adcrdy: out std_logic;
adcvalue: out std_logic_vector(13 downto 0);
mdrp_rdata: out std_logic_vector(7 downto 0);
vsenctl: in std_logic_vector(2 downto 0);
adcen: in std_logic;
clk: in std_logic;
drstn: in std_logic;
adcreqi: in std_logic;
adcmode: in std_logic;
mdrp_clk: in std_logic;
```

```
mdrp_wdata: in std_logic_vector(7 downto 0);
mdrp_a_inc: in std_logic;
mdrp_opcode: in std_logic_vector(1 downto 0)
);
end component;
Gowin_ADC_inst: Gowin_ADC
port map (
adcrdy => adcrdy_o,
adcvalue => adcvalue_o,
mdrp_rdata => mdrp_rdata_o,
vsenctl => vsenctl_i,
adcen => adcen_i,
clk => clk_i,
drstn => drstn_i,
adcreqi => adcreqi_i,
adcmode => adcmode_i,
mdrp_clk => mdrp_clk_i,
mdrp_wdata => mdrp_wdata_i,
mdrp_a_inc => mdrp_a_inc_i,
mdrp_opcode => mdrp_opcode_i
);
```

4 ADC (75K/138K)

4.1 Devices Supported

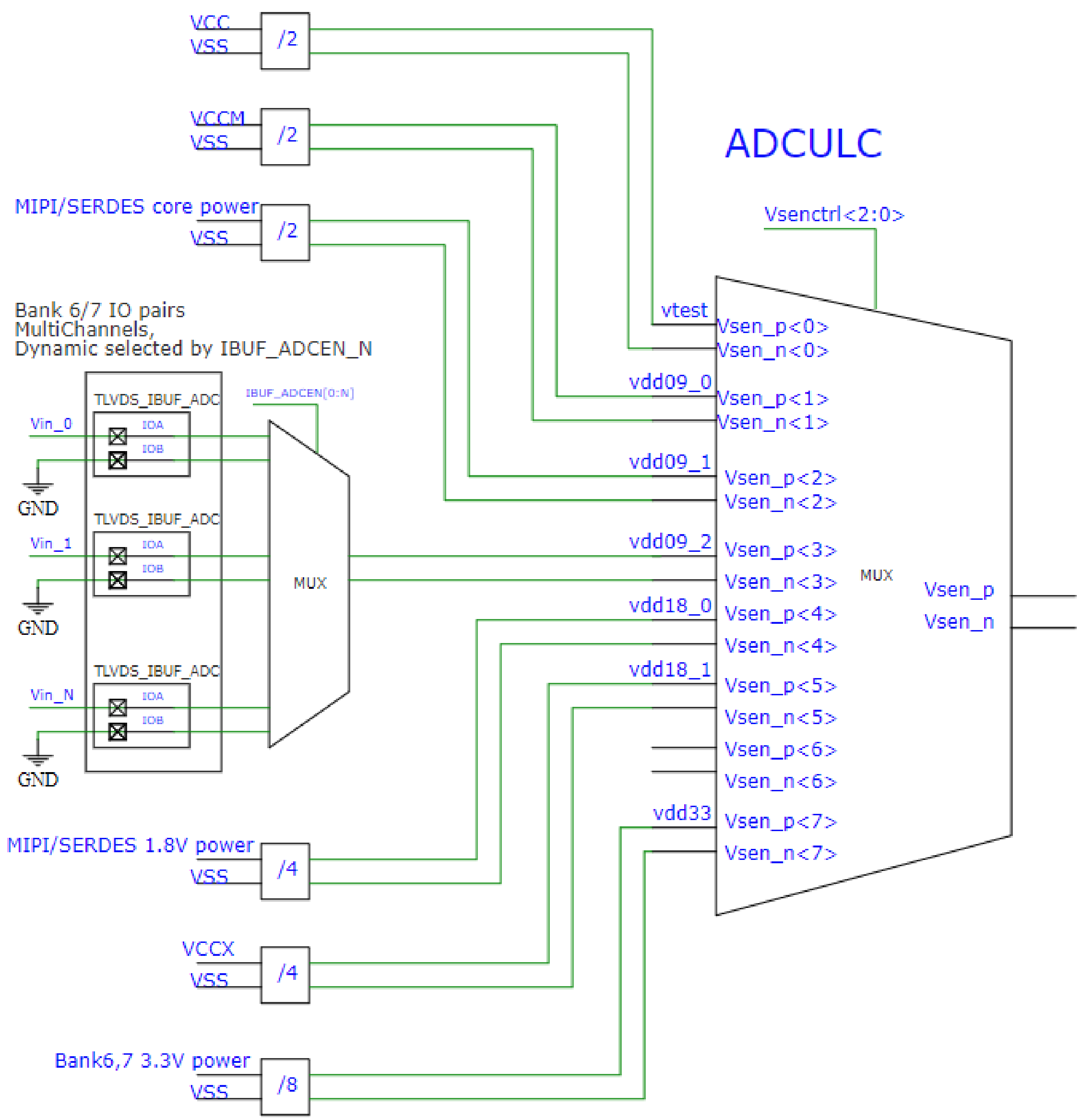
Table 4-1 Devices Supported

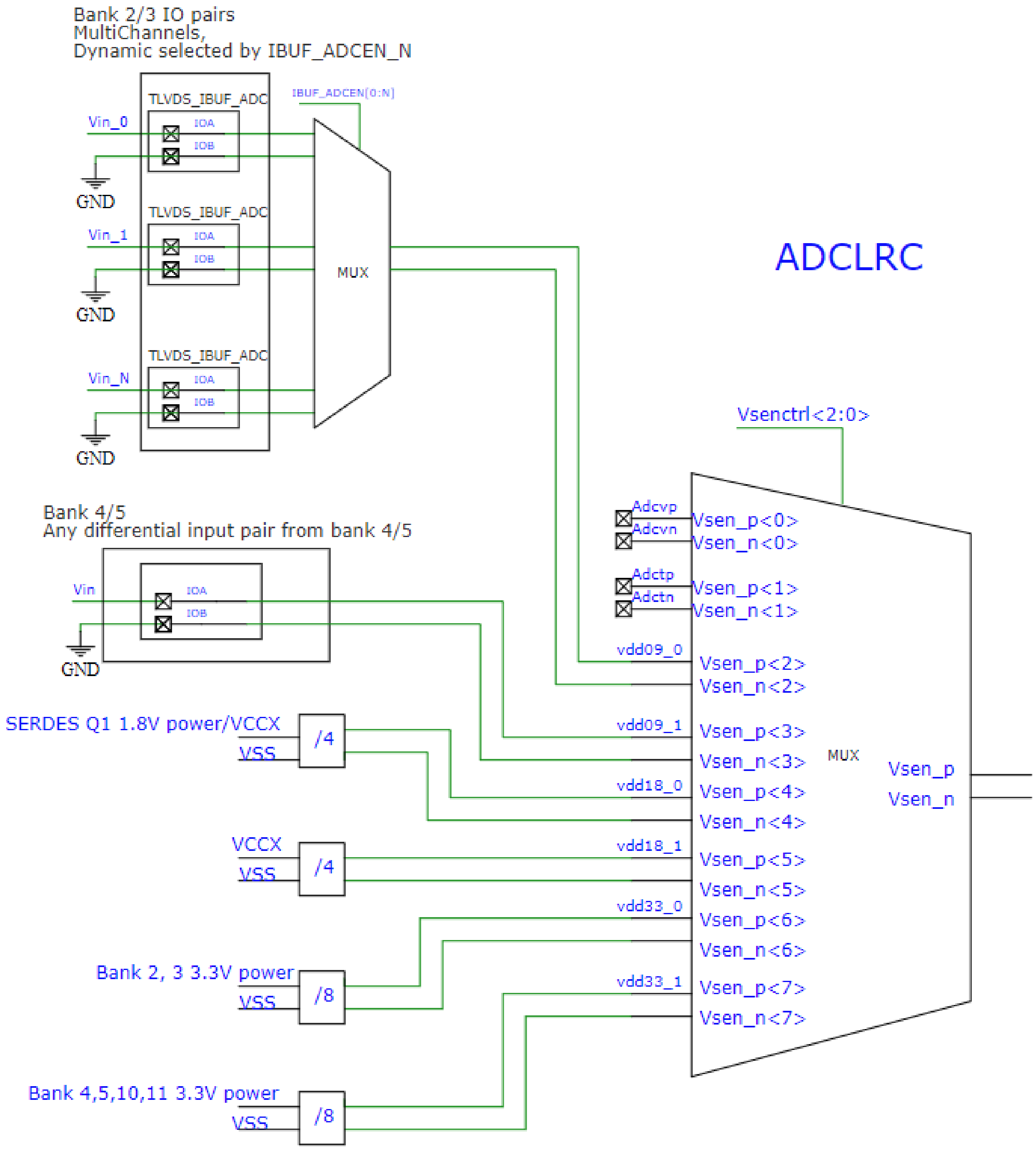
Product Family	Series	Device
Arora	GW5A	GW5A-138B
	GW5AS	GW5AS-138B
	GW5AT	GW5AT-138 / GW5AT-138B/ GW5AT-75B
	GW5AST	GW5AT-138B

4.2 Input Channel Selection (Voltage Divider Ratio)

The input channel selection diagram for voltage mode ADC in GW5A-138/GW5AT-138/ GW5AT-75/GW5AST-138 devices is shown below.

Figure 4-1 ADC Input Channel Selection Diagram in Voltage Mode





The input channel selection and voltage divider ratio table for voltage mode ADC in GW5A-138/GW5AT-138/GW5AT-75/GW5AST-138 devices is shown below.

Table 4-2 ADCULC Input Channel Selection and Divider Ratio in Voltage Mode

Vsenctl[2:0]	ADC Bus	Input Channel	Division Factor	Note
000	vtest	VCC	2	
001	vdd09_0	VCCM	2	

Vsenctl[2:0]	ADC Bus	Input Channel	Division Factor	Note
010	vdd09_1	VDDA_Q0	2	
		VDDT_Q0	2	
		VDDA_MIPI	2	
		VDDD_MIPI	2	
011	vdd09_2	ADCINBK6	1	BANK6 IO
		VCC	2	
		ADCINBK7	1	BANK7 IO
100	vdd18_0	VDDHA_Q0	4	
		VDDX_MIPI	4	
101	vdd18_1	VCCX	4	
111	vdd33	VCCIO6	8	
		VCCIO7	8	

Table 4-3 ADCLRC Input Channel Selection and Divider Ratio in Voltage Mode

Vsenctl[2:0]	ADC Bus	Input Channel	Division Factor	Note
000	adcv	adcv	1	
001	adct	adct	1	
010	vdd09_0	VDDA_Q1	2	
		VDDT_Q1	2	
		VCC	2	
		ADCINBK2	1	BANK2 IO
		ADCINBK3	1	BANK3 IO
011	vdd09_1	ADCINBK4	1	BANK4 IO
		VCC	2	
		ADCINBK5	1	BANK5 IO
100	vdd18_0	VDDHA_Q1	4	
		VCCX	4	
101	vcc18_1	VCCX	4	
110	vdd33_0	VCCIO2	8	
		VCCIO3	8	
111	vdd33_1	VCCIO4	8	
		VCCIO5	8	
		VCCIO10	8	

4.3 ADC Instantiation

4.3.1 Port Diagram

Figure 4-3 ADC Port Diagram (Static Input Channel)

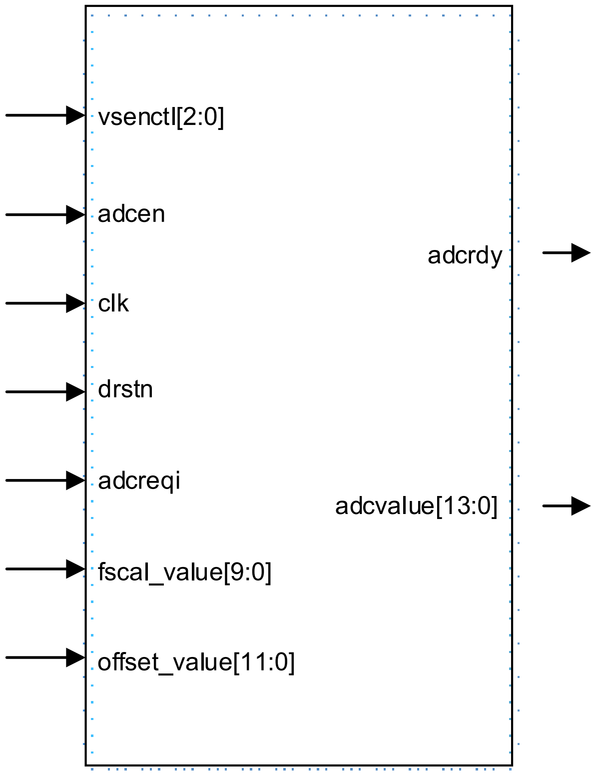
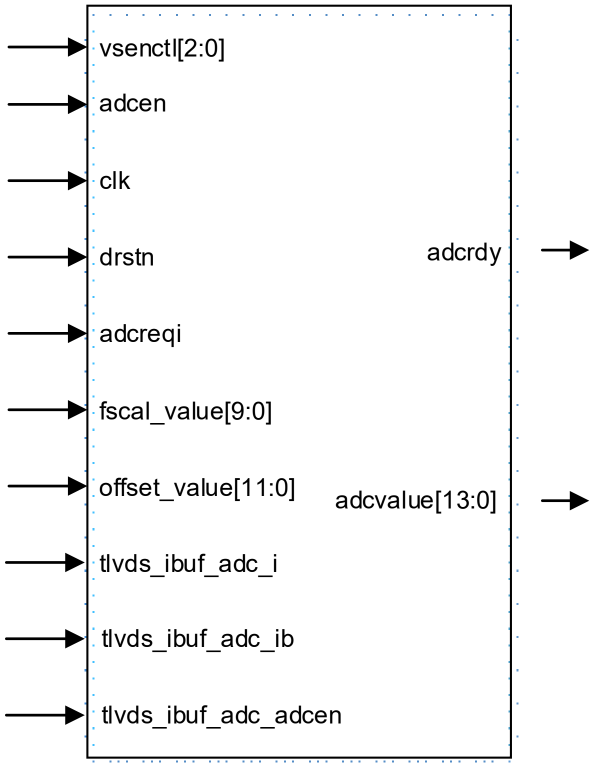


Figure 4-4 ADC Port Diagram (Dynamic Input Channel)



4.3.2 Port Description

ADCULC Port

Table 4-4 ADCULC Port Description

Port	I/O	Description
clk	input	clk input
drstn	input	digital part reset signal, active low
vsenctl	input	input source selection bit[2:0] ● 3'b000:vtest ● 3'b001:vdd09_0 ● 3'b010:vdd09_1 ● 3'b011:vdd09_2 ● 3'b100:vdd18_0 ● 3'b101:vdd18_1 ● 3'b111:vdd33
adcen	input	enable signal, active high
adcreqi	input	measurement request signal, valid rising edge
adcrdy	output	measurement completion signal, active high
adcvalue	output	bit[13:0] the measurement result output

Port	I/O	Description
fscal_value	input	bit[9:0] ● temperature mode: 510~948 ● voltage mode: 452~840
offset_value	input	bit[11:0] ● temperature mode: -1560~-760 ● voltage mode: -410~410
adcinbk6a	input	adcvp from the port A of Bank6 GPIO, with pin A connected to the voltage to be tested.
adcinbk6b	input	adcvn from the port B of Bank6 GPIO, with pin B tied to GND.
adcinbk7a	input	adcvp from the port A of Bank7 GPIO, with pin A connected to the voltage to be tested.
adcinbk7b	input	adcvn from the port B of Bank7 GPIO, with pin B tied to GND.

ADCULC_IO_BUF Port

Table 4-5 ADCULC_IO_BUF Port Description

Port	I/O	Description
tlvds_ibuf_adc_i	input	The adcvp signal is input to the differential GPIO Port A on Bank6/7, with Pin A connected to the voltage to be tested.
tlvds_ibuf_adc_ib	input	The adcvn signal is input to the differential GPIO Port B on Bank6/7, with Pin B tied to GND.
tlvds_ibuf_adc_adcen	input	The ADC enable signal for differential GPIOs; only one adcen can be enabled at a time. For Bank 6/7, only adcen needs to be controlled under the same vsenctrl.

Note!

Multiple pairs of IOs can be selected for testing on Bank6/7.

ADCLRC Port

Table 4-6 ADCLRC Port Description

Port	I/O	Description
clk	input	clk input
drstn	input	digital part reset signal, active low
vsenctl	input	input source selection bit[2:0] 3’b000: adcv 3’b001: adct 3’b010: vdd09_0

Port	I/O	Description
vsenctl	input	3’b011: vdd09_1 3’b100: vdd18_0 3’b101: vdd18_1 3’b110: vdd33_0 3’b111: vdd33_1
adcen	input	enable signal, active high
adcreqi	input	measurement request signal, valid rising edge
adcrdy	output	measurement completion signal, active high
adcvalue	output	bit[13:0] the measurement result output
fscal_value	input	bit[9:0] temperature mode: 510~948 voltage mode: 452~840
offset_value	input	bit[11:0] emperature mode: -1560~-760 voltage mode: -410~410
adcinbk2a	input	adcvp from Bank2 GPIO Pin A, with Pin A connected to the voltage to be tested.
adcinbk2b	input	adcvn from Bank2 GPIO Pin B, with Pin B tied to GND.
adcinbk3a	input	adcvp from Bank3 GPIO Pin A, with Pin A connected to the voltage to be tested.
adcinbk3b	input	adcvn from Bank3 GPIO Pin B, with Pin B tied to GND.
adcinbk4a	input	adcvp from Bank4 GPIO Pin A, with Pin A connected to the voltage to be tested.
adcinbk4b	input	adcvn from Bank4 GPIO Pin B, with Pin B tied to GND.
adcinbk5a	input	adcvp from Bank5 GPIO Pin A, with Pin A connected to the voltage to be tested.
adcinbk5b	input	adcvn from Bank5 GPIO Pin B, with Pin B tied to GND.

ADCLRC_IO_BUF Port

Table 4-7 ADCLRC_IO_BUF Port Description

Port	I/O	Description
tlvds_ibuf_adc_i	input	The adcvp signal is input to the differential GPIO Pin A on Bank2/3, with Pin A connected to the voltage to be tested.
tlvds_ibuf_adc_ib	input	The adcvn signal is input to the differential GPIO Pin B on Bank2/3, with Pin B tied to GND.

Port	I/O	Description
tlvds_ibuf_adc_adcen	input	The ADC enable signal for differential GPIOs; only one adcen can be enabled at a time. For Bank 2/3, only adcen needs to be controlled under the same vsenctrl.

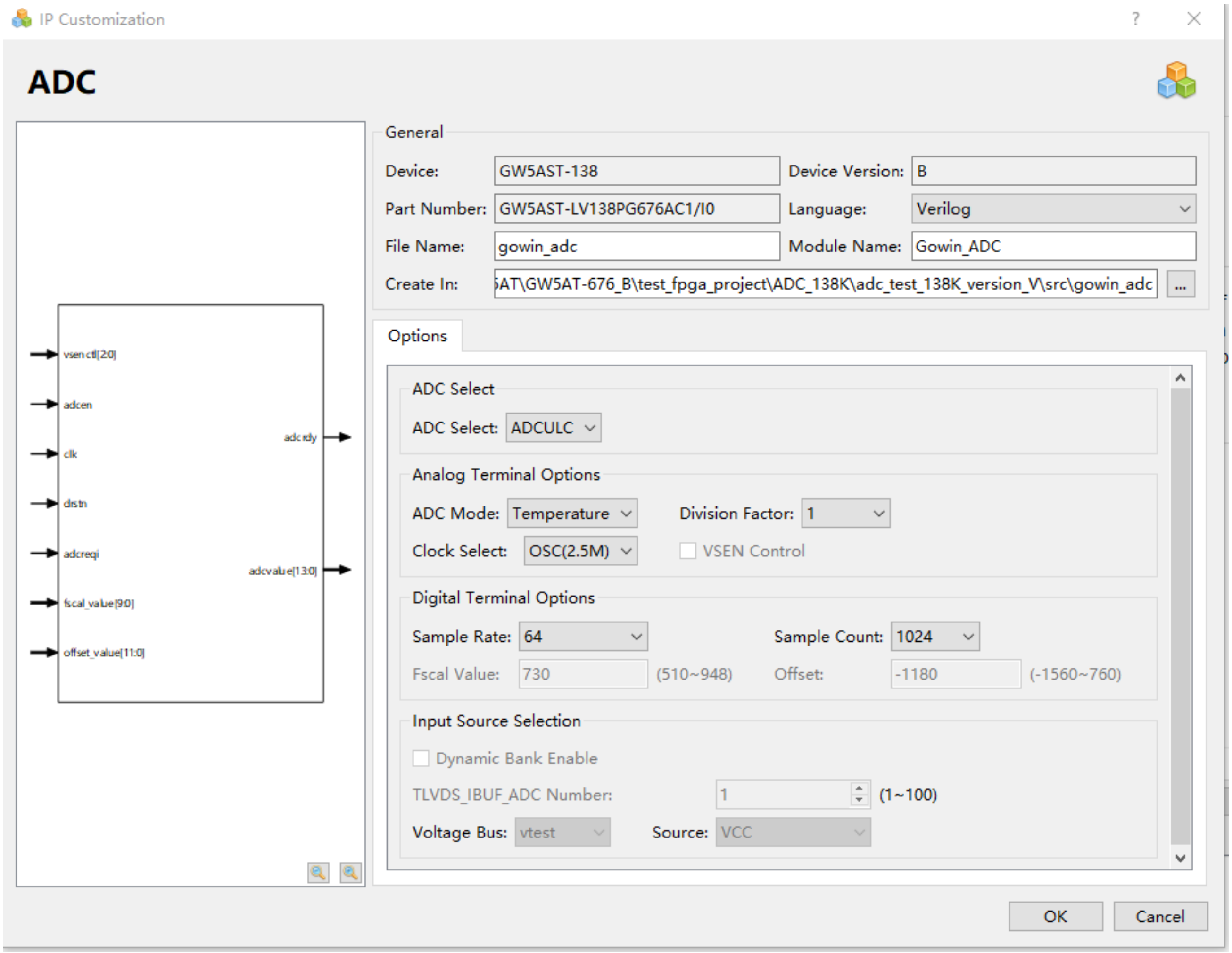
Note!

Multiple pairs of IOs can be selected for testing on Bank2/3.

4.3.3 Configuration Parameter Description

Select “Tools > IP Core Generator” in Gowin Software GUI to start the ADC configuration interface. Figure 4-5 shows the example.

Figure 4-5 ADC Configuration Interface



ADCULC Configuration Parameters

Table 4-8 ADCULC Configuration Parameter Description

Parameter	Default Value	Description
ADC Select	ADCULC	ADCULC/ADCLRC
ADC Mode	Temperature	Temperature/Voltage

Parameter	Default Value	Description
Division Factor	1	clock division 0: /1, 1: /2, 2: /4, 3: /8 Clock after frequency division, 500kHz~8MHz
Clock Select	OSC	clk source OSC(2.5MHz) /CLK/IO
VSEN Control	Unchecked	vsenctl port control The ADC IP generated upon selection does not include the vsenctl signal.
Sample Rate	128	sample rate configuration 16/32/64/128
Sample Count	1024	sample count configuration 128/256/512/1024
Fscal Value	730(Temperature) 653(Voltage)	temperature mode: 510~948 voltage mode: 452~840
Offset	-1180(Temperature) 0(Voltage)	temperature mode: -1560~-760 voltage mode: -410~410
Dynamic Input Source Enable (Voltage mode)	Unchecked	Checking the option enables the tlvds_ibuf_adc ADC input interface (Bank6/7 inputs, support multiple IO pairs)
vtest (Voltage mode)	VCC	VCC
vdd09_0 (Voltage mode)	VCCM	VCCM
vdd09_1 (Voltage mode)	VDDA_Q0	VDDA_Q0/VDDT_Q0/VDDA_MIPI/VDDD_MIPI
vdd09_2 (Voltage mode)	ADCINBK6	ADCINBK6/VCC/ ADCINBK7
Vdd18_0 (Voltage mode)	VDDHA_Q0	VDDHA_Q0/VDDX_MIPI
Vdd18_1 (Voltage mode)	VCCX	VCCX
Vdd33 (Voltage mode)	VCCIO6	VCCIO6/VCCIO7

ADCLRC Configuration Parameters

Table 4-9 ADCLRC Configuration Parameters Description

Parameter	Default Value	Description
ADC Select	ADCULC	ADCULC/ADCLRC
ADC Mode	Temperature	Temperature/Voltage
Division Factor	1	clock division 0: /1, 1: /2, 2: /4, 3: /8 Clock after frequency division, 500kHz~8MHz
Clock Select	OSC	clk source OSC(2.5MHz) /CLK/IO
VSEN Control	Unchecked	vsenctl control port When selected, the generated ADC IP does not include the vsenctl signal.
Sample Rate	128	sample rate configuration 16/32/64/128
Sample Count	1024	sample count configuration 128/256/512/1024
Fscal Value	730(Temperature) 653(Voltage)	temperature mode: 510~948 voltage mode: 452~840
Offset	-1180(Temperature) 0(Voltage)	temperature mode: -1560~-760 voltage mode: -410~410
Dynamic Input Source Enable (Voltage mode)	Unchecked	Checking the option enables the TLVDS_IBUF_ADC ADC input interface (Bank2/3 inputs, support multiple IO pairs)
vdd09_0 (Voltage mode)	VDDA_Q1	VDDA_Q1/VDDT_Q1/VCC/ADCINBK2/ADCINBK3
vdd09_1 (Voltage mode)	ADCINBK4	ADCINBK4/VCC/ ADCINBK5
vdd18_0 (Voltage mode)	VDDHA_Q1	VDDHA_Q1/VCCX
vdd18_1 (Voltage mode)	VCCX	VCCX
vdd33_0 (Voltage mode)	VCCIO2	VCCIO2/VCCIO3
vdd33_1	VCCIO4	VCCIO4/VCCIO5/VCCIO10

Parameter	Default Value	Description
(Voltage mode)	VCCIO4	VCCIO4/VCCIO5/VCCIO10

After ADC configuration, it will generate three files that are named after the "File Name". Take the default configuration as an example:

- “gowin_adc.v” file is a complete Verilog module to generate instance Gowin_ADC;
- “gowin_adc_tmp.v” is a template file for IP designs;
- “gowin_adc.ipc” file is an IP configuration file for users to load and configure the IP.

Note!

If VHDL is selected as the hardware description language, the first two files will be named with .vhd suffix.

4.3.4 Instantiation(ADCULC as an Example)

Verilog Instantiation:

```
Gowin_ADC Gowin_ADC_inst(
.adcrdy(adcrdy_o), //output adcrdy
.adcvalue(adcvalue_o), //output [13:0] adcvalue
.adcinbk6a(adcinbk6a_i), //input adcinbk6a
.adcinbk6b(adcinbk6b_i), //input adcinbk6b
.adcinbk7a(adcinbk7a_i), //input adcinbk7a
.adcinbk7b(adcinbk7b_i), //input adcinbk7b
.vsenctl(vsenctl_i), //input [2:0] vsenctl
.adcen(adcen_i), //input adcen
.clk(clk_i), //input clk
.drstn(drstn_i), //input drstn
.adcreqi(adcreqi_i) ,//input adcreqi
.fscal_value(fscal_value_i), //input [9:0] fscal_value
.offset_value(offset_value_i) //input [11:0] offset_value
);
```

Vhdl Instantiation:

```
component Gowin_ADC
port (
adcrdy: out std_logic;
adcvalue: out std_logic_vector(13 downto 0);
adcinbk6a: in std_logic;
adcinbk6b: in std_logic;
adcinbk7a: in std_logic;
adcinbk7b: in std_logic;
vsenctl: in std_logic_vector(2 downto 0);
```

```
adcen: in std_logic;
clk: in std_logic;
drstn: in std_logic;
adcreqi: in std_logic;
fscal_value: in std_logic_vector(9 downto 0);
offset_value: in std_logic_vector(11 downto 0)
);
end component;
Gowin_ADC_inst: Gowin_ADC
port map (
adcrdy => adcrdy_o,
adcvalue => adcvalue_o,
adcinbk6a => adcinbk6a_i,
adcinbk6b => adcinbk6b_i,
adcinbk7a => adcinbk7a_i,
adcinbk7b => adcinbk7b_i,
vsenctl => vsenctl_i,
adcen => adcen_i,
clk => clk_i,
drstn => drstn_i,
adcreqi => adcreqi_i,
fscal_value => fscal_value_i,
offset_value => offset_value_i
);
```

5 ADC (60K)

5.1 Devices Supported

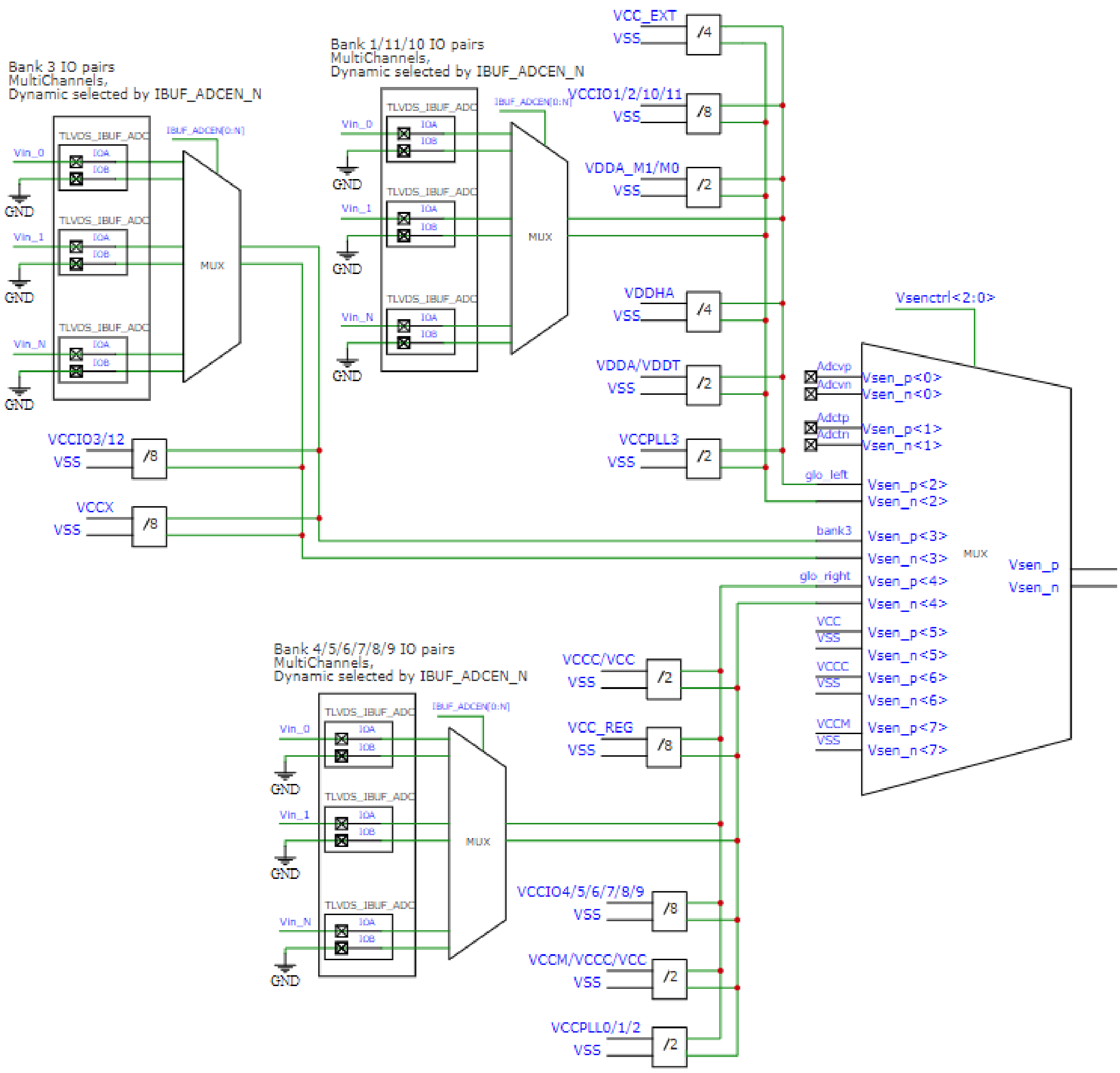
Table 5-1 Devices Supported

Product Family	Series	Device
Arora	GW5A	GW5A-60
	GW5AT	GW5AT-60

5.2 Input Channel Selection (Voltage Divider Ratio)

The input channel selection diagram for voltage mode ADC in GW5A-60/GW5AT-60 devices is shown below.

Figure 5-1 ADC Input Channel Selection Diagram in Voltage Mode



The input channel selection and voltage divider ratio table for voltage mode ADC in GW5A-60/GW5AT-60 devices is shown below.

Table 5-2 ADC Input Channel Selection and Voltage Divider Ratio in Voltage Mode

Vsenctl[2:0]	ADC Bus	Input Channel	Division Factor	BUF_EN[28:0]	Note
000	adcv		1		

Vsenctl[2:0]	ADC Bus	Input Channel	Division Factor	BUF_EN[28:0]	Note
001	adct		1		
010	glo_left	VDD12_MIPI	4	BUF_EN[0]	VCC_EXT
		VCCIO1	8	BUF_EN[1]	
		VCCIO2	8	BUF_EN[2]	
		VCCIO10	8	BUF_EN[15]	
		VCCIO11	8	BUF_EN[16]	
		VDDA_MIPI	2	BUF_EN[20]	VDDA_M1/M0
		VDDHA_Q0	4	BUF_EN[22]	VDDHA
		VDDA_Q0	2	BUF_EN[23]	VDDA
		VDDT_Q0	2	BUF_EN[24]	VDDT
		VCCPLL3	2	BUF_EN[28]	
		PAD	1		BANK1/11/10 上 IO
011	bank3	VCCIO3	8	BUF_EN[3]	
		VCCIO12	8	BUF_EN[4]	
		VCCX	8	BUF_EN[5]	
		PAD	1		BANK3 IO
100	glo_right	VCC	2	BUF_EN[7]	
		VCC_LDO	8	BUF_EN[8]	VCC_REG
		VCCIO4	8	BUF_EN[9]	
		VCCIO5	8	BUF_EN[10]	
		VCCIO6	8	BUF_EN[11]	
		VCCIO7	8	BUF_EN[12]	
		VCCIO8	8	BUF_EN[13]	
		VCCIO9	8	BUF_EN[14]	
		VCCM	2	BUF_EN[17]	
		VCCPLL0	2	BUF_EN[25]	
		VCCPLL1	2	BUF_EN[26]	
		VCCPLL2	2	BUF_EN[27]	
		PAD	1		BANK4/5/6/7/8/9 IO
101	vcc		1		
110	vccc		1		
100	vccm		1		

5.3 ADC Instantiation

5.3.1 Port Diagram

Figure 5-2 ADC Port Diagram (Static Input Channel)

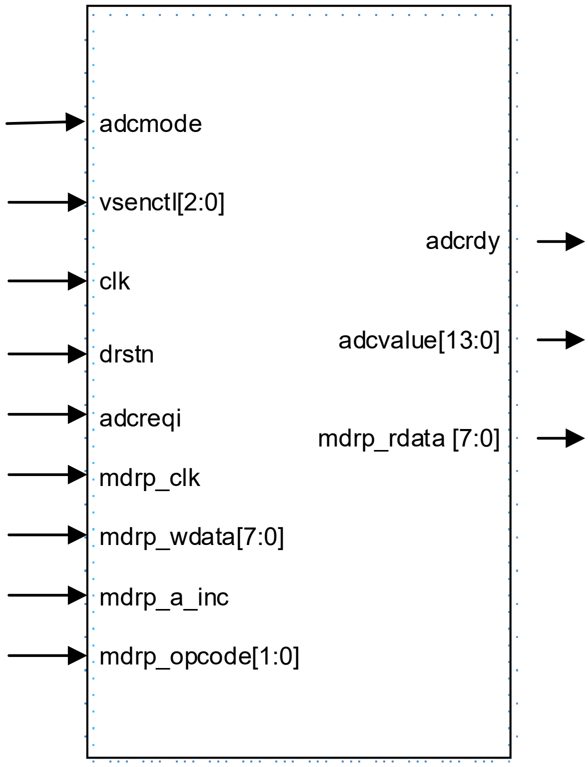
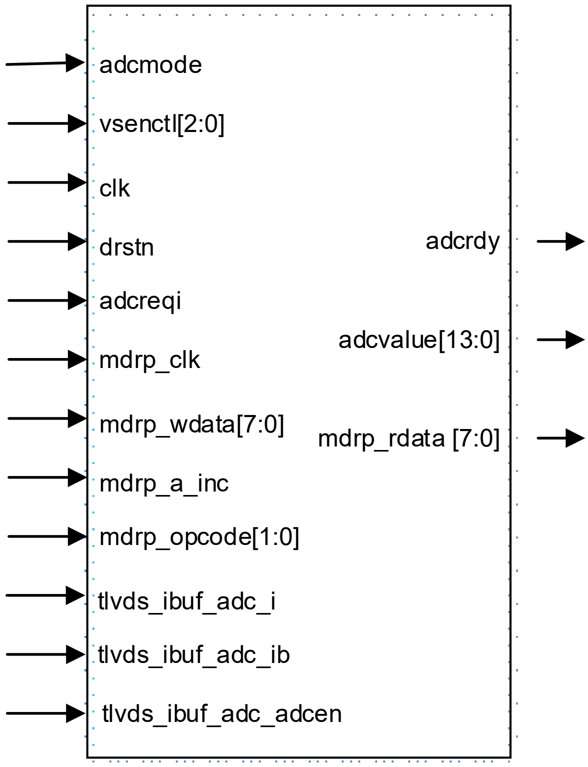


Figure 5-3 ADC Port Diagram (Dynamic Input Channel)



5.3.2 Port Description

ADC Port

Table 5-3 ADC Port Description

Port	I/O	Description
adcmode	input	mode selection ● 1'b0: temperature mode ● 1'b1:voltage mode
vsenctl[2:0]	input	input source selection bit [2:0] ● 3'b000: adcv ● 3'b001: adct ● 3'b010: glo_left ● 3'b011: bank3 ● 3'b100: glo_right ● 3'b101: vcc ● 3'b110: vccc ● 3'b111: vccm
clk	input	clk input
drstn	input	digital part reset signal, active low
adcreqi	input	measurement request signal, valid rising edge

Port	I/O	Description
adcrdy	output	measurement completion signal, active high
adcvalue	output	bit[13:0] the measurement result output

ADC_mDRP Port

Table 5-4 ADC_MDRP Port Description

Port	I/O	Description
mdrp_clk	input	mdrp clock
mdrp_wdata[7:0]	input	bit[7:0] mdrp_wdata
mdrp_a_inc	input	mdrp_a_inc
mdrp_opcode[1:0]	input	bit[1:0] mdrp_opcode
mdrp_rdata	output	bit[7:0] mdrp_rdata

mDRP Control Registers

Table 5-5 ADC mDRP Control Registers

Address	Register name	R/W	Description
0x00	sensor_adc_ctrl1	RW	Bit[7:6] reserved Bit[5:3] cfg_sample_cnt_sel ● 001:128 ● 010:256 ● 011:512 ● 100:1024 Bit[2:0] gain_ctrl: Keep at 0.
0x01	sensor_adc_ctrl2	RW	Bit[7:4] reserved Bit[3] cfg_sensor_mode ● 0: Temperature ● 1:Voltage Bit[2:0] cfg_rate_change_ctrl ● 010:16 ● 011:32 ● 100:64 ● 101:128
0x02	sensor_adc_ctrl3	RW	Bit[7:0] cfg_fscal_value[7:0]
0x03	sensor_adc_ctrl4	RW	● Bit[7:2] reserved ● Bit[1:0] cfg_fscal_value[9:8]
0x04	sensor_adc_ctrl5	RW	Bit[7:0] cfg_offset_value[7:0]
0x05	sensor_adc_ctrl6	RW	● Bit[7:4] reserved

Address	Register name	R/W	Description
0x05	sensor_adc_ctrl6	RW	• Bit[3:0] cfg_offset_value[11:8]

ADC_IO_BUF Port

Table 5-6 ADC_IO_BUF Port Description

Port	I/O	Description
tlvds_ibuf_adc_i	input	The adcvp signal is input to the differential GPIO Pin A, with Pin A connected to the voltage to be tested.
tlvds_ibuf_adc_ib	input	The adcvn signal is input to the differential GPIO Pin B, with Pin B tied to GND.
tlvds_ibuf_adc_adcen	input	The ADC enable signal for differential GPIOs; only one adcen can be enabled at a time. Dynamic control is achieved through the combination of vsenc-trl and ad-cen. **Note!** Ensure that all PADs are selected in the ADC configura-tion page.

Note!

Multiple pairs of IOs can be selected for testing on all Banks.

5.3.3 Configuration Parameter Description

Select “Tools > IP Core Generator” in Gowin Software GUI to start the ADC configuration interface. Figure 5-4 shows the example.

Figure 5-4 ADC Configuration Interface

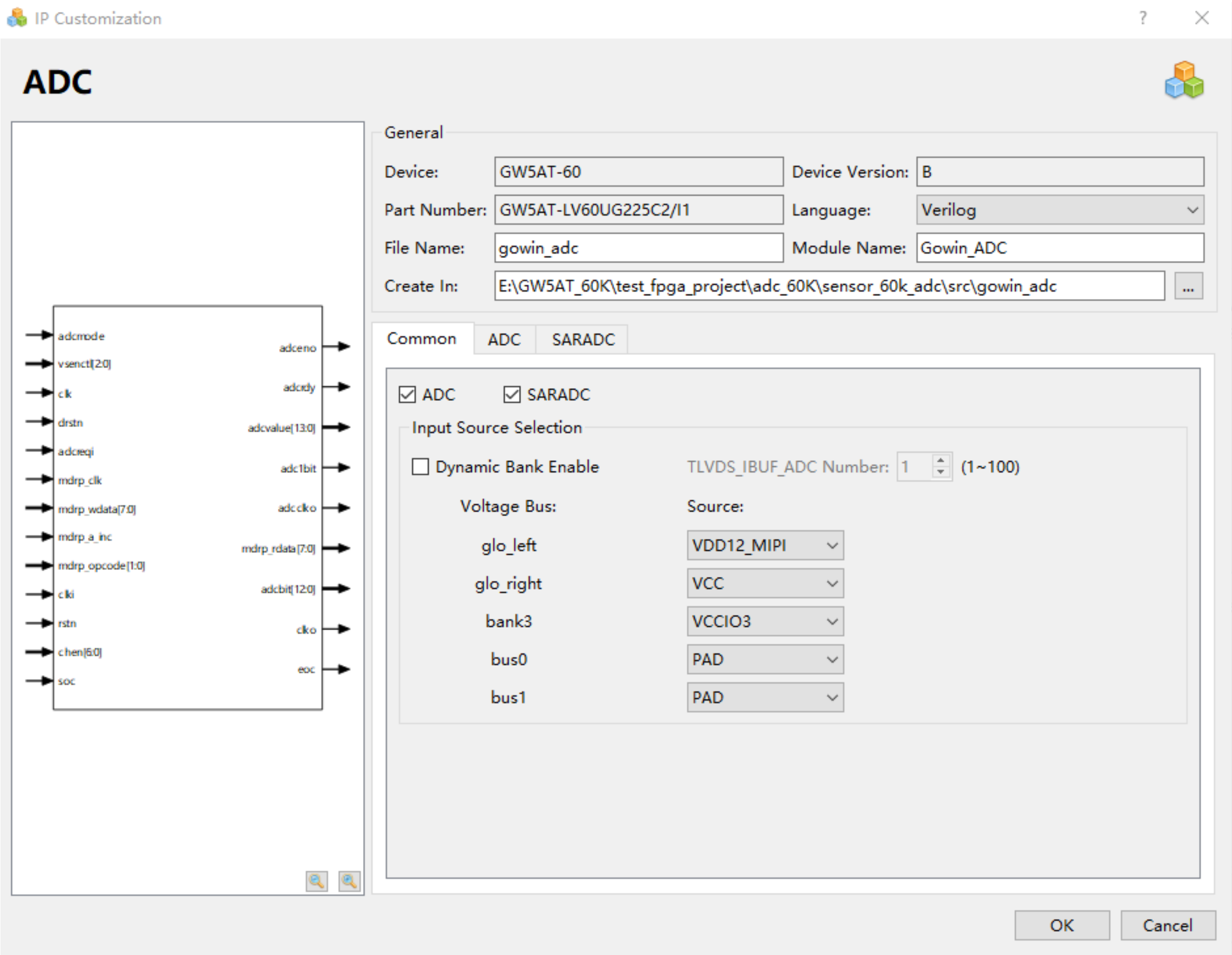


Table 5-7 ADC Configuration Parameter Description

Parameter	Default Value	Description
Common	ADC/SARADC	ADC
ADC Mode	Voltage	Temperature/Voltage
ADC Enable Selection	Unchecked	If not selected, the ADC remains continuously active. If selected, the adceni signal is instantiated, allowing external signals to control the ADC's operating state.
Clock Select	CLK	clk source OSC (2.5MHz) or CLK
Division Factor	1	clock division 0: /1, 1: /2, 2: /4, 3: /8 Clock after frequency division, 500kHz~8MHz
Sample Rate	128	sample rate configuration 16/32/64/128
Sample Count	1024	sample count configuration

Parameter	Default Value	Description
Sample Count	1024	128/256/512/1024
Fscal Value	730(Temperature) 653(Voltage)	temperature mode: 510~948 voltage mode: 452~840
Offset	-1180(Temperature) 0(Voltage)	temperature mode: -1560~-760 voltage mode: -410~410
Dynamic Input Source Enable (Voltage mode)	Unchecked	When checked, the tlvds_ibuf_adc ADC input interface is enabled. (Input from all banks, supporting multiple IO pairs)
glo_left (Voltage mode)	VDD12_MIPI	VDD12_MIPI/VCCIO1_2_10_11/VDDA_MIPI/VDDHA_Q0/VDDA_Q0/VDDT_Q0/VCCPLL3/PAD PAD corresponds to IOs in Bank1/10/11. When using IOs from these Banks as ADC inputs without selecting Dynamic Input Source Enable, the following physical constraint must be added: USE_ADC_SRC bus2 loc; loc: Specifies the physical location of the ADC input pin, e.g., IOR26. When Dynamic Input Source Enable is selected, the tlvds_ibuf_adc ADC input interface is enabled. Connect terminal A to the voltage under test, and terminal B to ground (GND).
glo_right (Voltage mode)	VCC	VCC/VCC_LDO/VCCIO4_5_6_7_8_9/VCCM/VCCPLL0_1_2 / PAD PAD corresponds to IOs in Bank4/5/6/7/8/9. When using IOs from these Banks as ADC inputs without selecting Dynamic Input Source Enable, the following physical constraint must be added: USE_ADC_SRC bus4 loc; loc: Specifies the physical location of the ADC input pin, e.g., IOR26. When Dynamic Input Source Enable is selected, the tlvds_ibuf_adc ADC input interface is enabled. Connect terminal A to the voltage under test, and terminal B to ground (GND).
bank3 (Voltage mode)	VCCIO3	VCCIO3/VCCIO12/VCCX/PAD PAD corresponds to IOs in Bank3. When using IOs from these Banks as ADC inputs without selecting Dynamic Input Source Enable, the following physical constraint must be added: USE_ADC_SRC bus3 loc;

Parameter	Default Value	Description
bank3 (Voltage mode)	VCCIO3	loc: Specifies the physical location of the ADC input pin, e.g., IOR26. When Dynamic Input Source Enable is selected, the tlvds_ibuf_adc ADC input interface is enabled. Connect terminal A to the voltage under test, and terminal B to ground (GND).
bus0	PAD	PAD (used in SARADC)
bus1	PAD	PAD (used in SARADC)

After ADC configuration, it will generate three files that are named after the "File Name". Take the default configuration as an example:

- “gowin_adc.v” file is a complete Verilog module to generate instance Gowin_ADC;
- “gowin_adc_tmp.v” is a template file for IP designs;
- “gowin_adc.ipc” file is an IP configuration file for users to load and configure the IP.

Note!

If VHDL is selected as the hardware description language, the first two files will be named with .vhd suffix.

5.3.4 Instantiation

Verilog Instantiation:

```
Gowin_ADC Gowin_ADC_inst (
.adceno(adceno), //output adceno
.adcrdy(adcrdy), //output adcrdy
.adcvalue(adcvalue), //output [13:0] adcvalue
.mdrp_rdata(mdrp_rdata), //output [7:0] mdrp_rdata
.adc1bit(adc1bit), //output adc1bit
.adcclko(adcclko), //output adcclko
.adcmode(adcmode), //input adcmode
.vsenctl(vsenctl), //input [2:0] vsenctl
.clk(clk), //input clk
.drstn(drstn), //input drstn
.adcreqi(adcreqi), //input adcreqi
.mdrp_clk(mdrp_clk), //input mdrp_clk
.mdrp_wdata(mdrp_wdata), //input [7:0] mdrp_wdata
.mdrp_a_inc(mdrp_a_inc), //input mdrp_a_inc
.mdrp_opcode(mdrp_opcode) //input [1:0] mdrp_opcode
);
```

Vhdl Instantiation:

```
component Gowin_ADC
port (
adceno: out std_logic;
adcrdy: out std_logic;
adcvalue: out std_logic_vector(13 downto 0);
mdrp_rdata: out std_logic_vector(7 downto 0);
adc1bit: out std_logic;
adcclko: out std_logic;
adcbit: out std_logic_vector(12 downto 0);
clko: out std_logic;
eoc: out std_logic;
adcmode: in std_logic;
vsenctl: in std_logic_vector(2 downto 0);
clk: in std_logic;
drstn: in std_logic;
adcreqi: in std_logic;
mdrp_clk: in std_logic;
mdrp_wdata: in std_logic_vector(7 downto 0);
mdrp_a_inc: in std_logic;
mdrp_opcode: in std_logic_vector(1 downto 0);
clki: in std_logic;
chen: in std_logic_vector(6 downto 0);
rstn: in std_logic;
soc: in std_logic
);
end component;
your_instance_name: Gowin_ADC
port map (
adceno => adceno,
adcrdy => adcrdy,
adcvalue => adcvalue,
mdrp_rdata => mdrp_rdata,
adc1bit => adc1bit,
adcclko => adcclko,
adcbit => adcbit,
clko => clko,
eoc => eoc,
adcmode => adcmode,
```

```
vsenctl => vsenctl,
clk => clk,
drstn => drstn,
adcreqi => adcreqi,
mdrp_clk => mdrp_clk,
mdrp_wdata => mdrp_wdata,
mdrp_a_inc => mdrp_a_inc,
mdrp_opcode => mdrp_opcode,
clki => clki,
chen => chen,
rstn => rstn,
soc => soc
);
```

6 ADC (15K)

6.1 Devices Supported

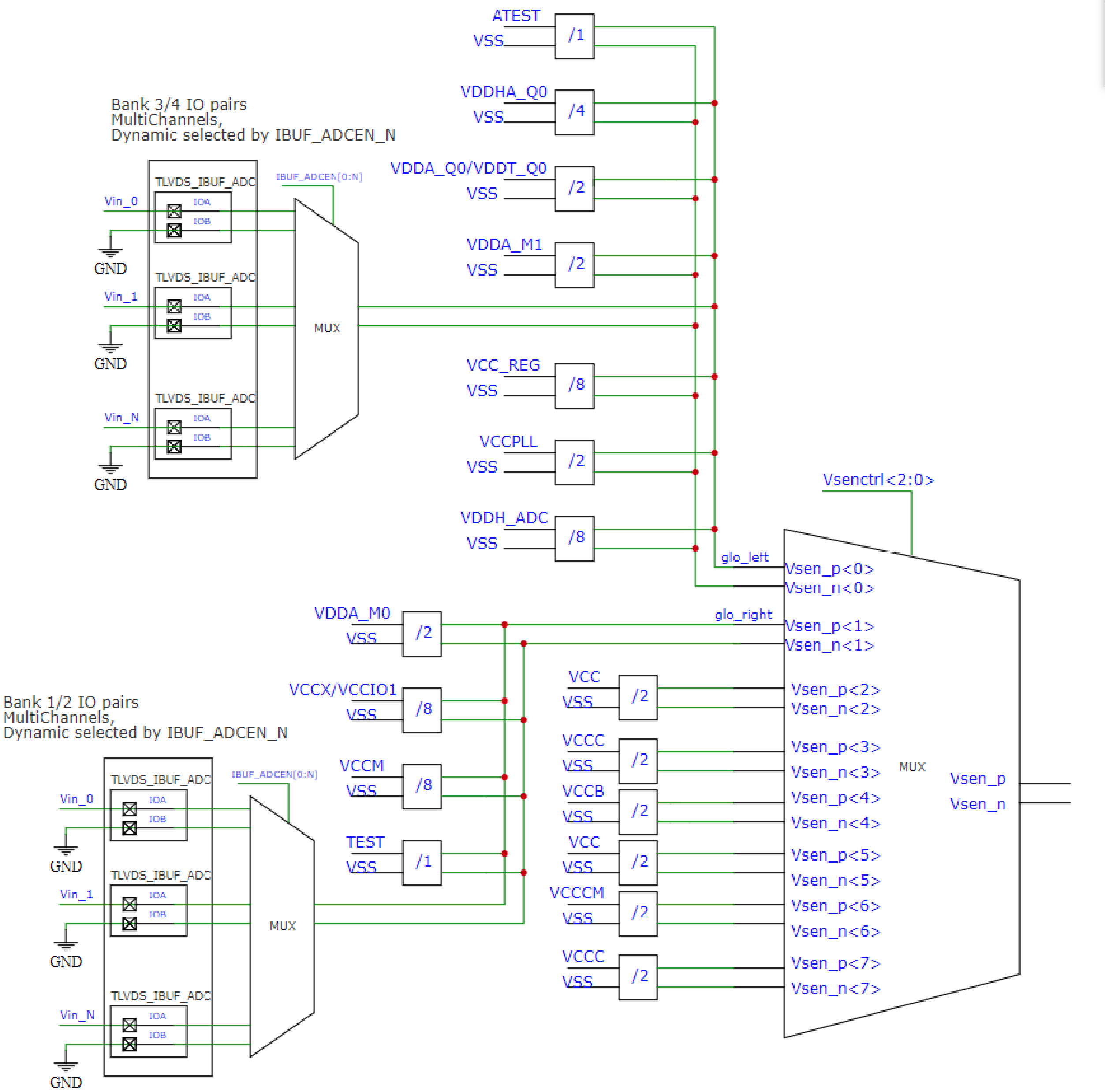
Table 6-1 Devices Supported

Product Family	Series	Device
Arora V	GW5AT	GW5AT-15
	GW5ART	GW5ART-15

6.2 Input Channel Selection (Voltage Divider Ratio)

The input channel selection diagram for voltage mode ADC in GW5AT-15/GW5ART-15 devices is shown below.

Figure 6-1 ADC Input Channel Selection Diagram in Voltage Mode



The input channel selection and voltage divider ratio table for voltage mode ADC in GW5AT-15/GW5ART-15 devices is shown below.

TBD

6.3 ADC Instantiation

6.3.1 Port Diagram

Figure 6-2 ADC Port Diagram (Static Input Channel)

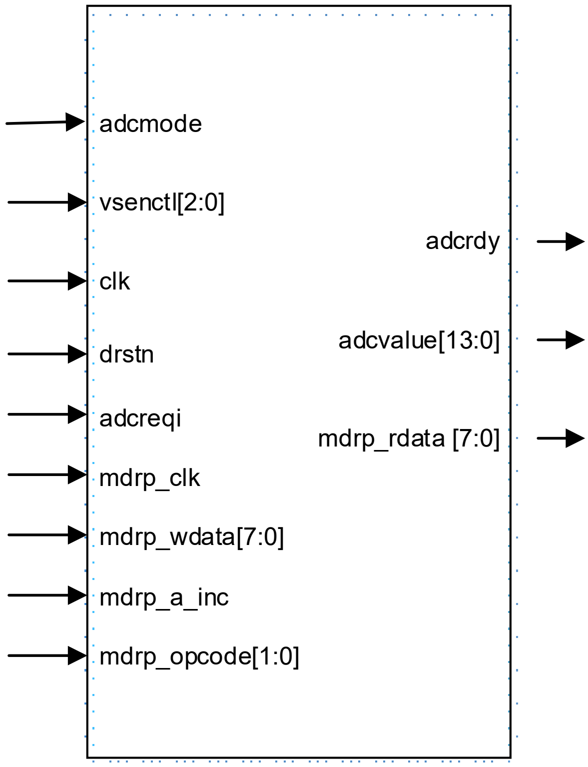
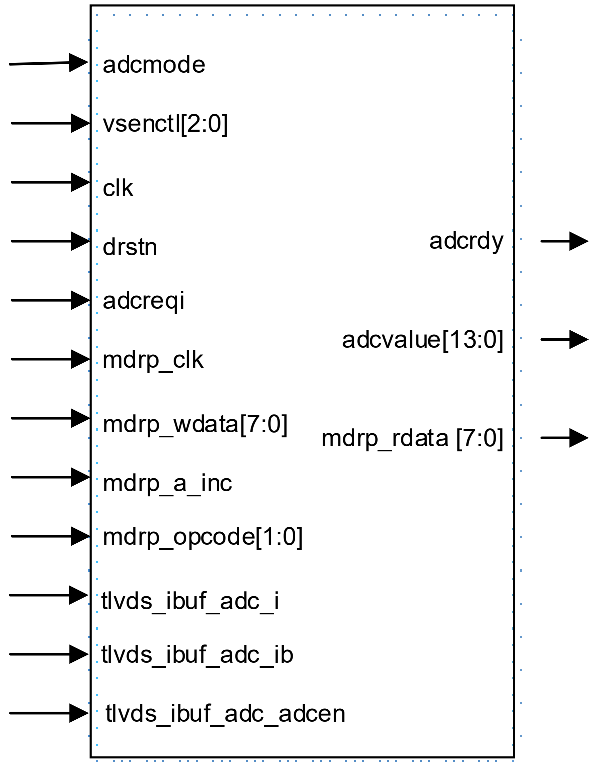


Figure 6-3 ADC Port Diagram (Dynamic Input Channel)



6.3.2 Port Description

ADC Port

Table 6-2 ADC Port Description

Port	I/O	Description
adcmode	input	mode selection ● 1'b0: temperature mode ● 1'b1: voltage mode
vsenctl[2:0]	input	input source selection bit [2:0] ● 3'b000: glo_left ● 3'b001: glo_right ● 3'b010: vcc ● 3'b011: vccc ● 3'b100: vccb ● 3'b101: vcc ● 3'b110: vccm ● 3'b111: vccc
clk	input	clk input
drstn	input	digital part reset signal, active low
adcreqi	input	measurement request signal, valid rising edge

Port	I/O	Description
adcrdy	output	measurement completion signal, active high
adcvalue	output	bit[13:0] the measurement result output

ADC_mDRP Port

Table 6-3 ADC_mDRP Port Description

Port	I/O	Description
mdrp_clk	input	mdrp clock
mdrp_wdata[7:0]	input	bit[7:0] mdrp_wdata
mdrp_a_inc	input	mdrp_a_inc
mdrp_opcode[1:0]	input	bit[1:0] mdrp_opcode
mdrp_rdata	output	bit[7:0] mdrp_rdata

mDRP Control Registers

Table 6-4 ADC mDRP Control Registers

Address	Register name	R/W	Description
0x00	sensor_adc_ctrl1	RW	Bit[7:6] reserved Bit[5:3] cfg_sample_cnt_sel ● 001:128 ● 010:256 ● 011:512 ● 100:1024 Bit[2:0] gain_ctrl: Keep at 0.
0x01	sensor_adc_ctrl2	RW	Bit[7:4] reserved Bit[3] cfg_sensor_mode ● 0: Temperature ● 1:Voltage Bit[2:0] cfg_rate_change_ctrl ● 010:16 ● 011:32 ● 100:64 ● 101:128
0x02	sensor_adc_ctrl3	RW	Bit[7:0] cfg_fscal_value[7:0]
0x03	sensor_adc_ctrl4	RW	● Bit[7:2] reserved ● Bit[1:0] cfg_fscal_value[9:8]
0x04	sensor_adc_ctrl5	RW	Bit[7:0] cfg_offset_value[7:0]
0x05	sensor_adc_ctrl6	RW	● Bit[7:4] reserved

Address	Register name	R/W	Description
0x05	sensor_adc_ctrl6	RW	● Bit[3:0] cfg_offset_value[11:8]

ADC_IO_BUF Port

Table 6-5 ADC_IO_BUF Port Description

Port	I/O	Description
tlvds_ibuf_adc_i	input	The adcvp signal is input to the differential GPIO Port A on Bank1/2/3/4, with Pin A connected to the voltage to be tested.
tlvds_ibuf_adc_ib	input	The adcvn signal is input to the differential GPIO Port B on Bank1/2/3/4, with Pin B tied to GND.
tlvds_ibuf_adc_adcen	input	The ADC enable signal for differential GPIOs; only one adcen can be enabled at a time. Dynamic control is achieved through the combination of vsenc-trl and ad-cen. **Note!** Ensure that all PADs are selected in the ADC configura-tion page.

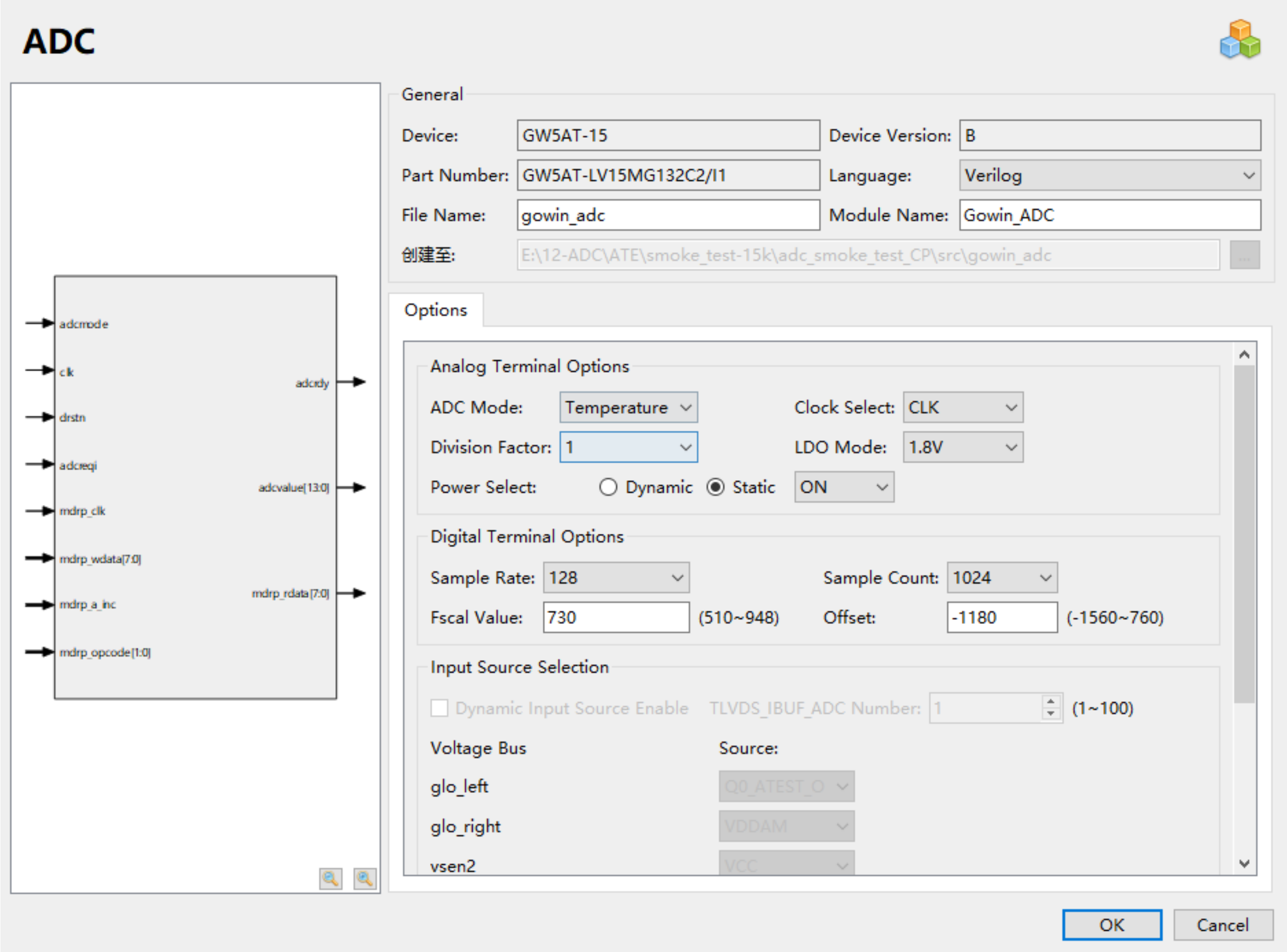
Note!

Multiple pairs of IOs can be selected for testing on Bank1/2/3/4.

6.3.3 Configuration Parameter Description

Select “Tools > IP Core Generator” in Gowin Software GUI to start the ADC configuration interface. Figure 6-4 shows the example.

Figure 6-4 ADC Configuration Interface



6.3.4 Instantiation

Verilog

```
Gowin_ADC Gowin_ADC_inst (
.adcrdy(adcrdy_o),
.adcvalue(adcvalue_o),
.mdrp_rdata(mdrp_rdata_o),
.vsenctl(vsenctl_i),
.adcen(adcen_i),
.clk(clk_i),
.drstn(drstn_i),
.adcreqi(adcreqi_i),
.adcmode(adcmode_i),
.mdrp_clk(mdrp_clk_i),
.mdrp_wdata(mdrp_wdata_i),
.mdrp_a_inc(mdrp_a_inc_i),
.mdrp_opcode(mdrp_opcode_i)
```

```
);
Vhdl
component Gowin_ADC
port (
adcrdy: out std_logic;
adcvalue: out std_logic_vector(13 downto 0);
mdrp_rdata: out std_logic_vector(7 downto 0);
vsenctl: in std_logic_vector(2 downto 0);
adcen: in std_logic;
clk: in std_logic;
drstn: in std_logic;
adcreqi: in std_logic;
adcmode: in std_logic;
mdrp_clk: in std_logic;
mdrp_wdata: in std_logic_vector(7 downto 0);
mdrp_a_inc: in std_logic;
mdrp_opcode: in std_logic_vector(1 downto 0)
);
end component;
Gowin_ADC_inst: Gowin_ADC
port map (
adcrdy => adcrdy_o,
adcvalue => adcvalue_o,
mdrp_rdata => mdrp_rdata_o,
vsenctl => vsenctl_i,
adcen => adcen_i,
clk => clk_i,
drstn => drstn_i,
adcreqi => adcreqi_i,
adcmode => adcmode_i,
mdrp_clk => mdrp_clk_i,
mdrp_wdata => mdrp_wdata_i,
mdrp_a_inc => mdrp_a_inc_i,
mdrp_opcode => mdrp_opcode_i
);
```

GOWIN

PROGRAMMING FOR THE FUTURE